DO NOW — RECALL

Lesson 054 · Hash Tables and Dictionary<TKey,TValue>

Name: _____ Class: _____ Date: _____

1. Which data structure uses FIFO in BFS?

2. What does a dictionary map?

3. What is a collision in hashing?

4. A hash table has slots 0–6, all empty. Insert keys 9, 16 and 23 in that order using h(k)=k MOD 7 and linear probing to the next slot. Determine the final slot occupied by 23.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

A queue.

2. Model answer

Keys to associated values.

3. Model answer

Different keys map to same hash location.

4. Model answer

Slot 4. All three hash to 2; 9 occupies 2, 16 occupies 3, and 23 probes to 4.

By the end, you can…

• A hash function maps keys to hash values that guide placement/lookups. A collision occurs if distinct keys map to the same location/bucket.

• A .NET Dictionary<TKey,TValue> handles collision management internally; do not equate its actual buckets with simplistic modulo examples.

• To read a missing key safely, use TryGetValue. Hash-table operations are often expected O(1) but poor collisions can degrade performance.

Vocabulary — know these terms

hash function — Maps keys to hash values.

collision — Different keys share a target bucket.

bucket — Hash table storage location.

load factor — Filled entries relative to capacity.

NEW LEARNING

Read, annotate and explain

Discover

A hash function maps keys to hash values that guide placement/lookups. A collision occurs if distinct keys map to the same location/bucket.

Trace

A .NET Dictionary<TKey,TValue> handles collision management internally; do not equate its actual buckets with simplistic modulo examples.

Evaluate

To read a missing key safely, use TryGetValue. Hash-table operations are often expected O(1) but poor collisions can degrade performance.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Apply a toy hash h(k)=k mod 5 to keys 7 and 12. What happens?

Explain the trace

Both keys collide in bucket 2; a table must resolve collision by chaining or probing.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

With toy h(k)=k mod5, find buckets for keys 8,13,19. [6]

Question 2

Write C# Dictionary<string,int> code storing marks Ada80 and Ben65, then safely test whether Cleo exists. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Compare chaining and linear probing for toy hash collisions, and state two performance concerns. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Hash tables can never have collisions when keys are different.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.